Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Thermal Care of the Newborn

Eastern & Northern Services

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SUMMARY

The purpose of this document is to detail the process for the thermal care of neonates.

Thermoregulation is the ability to maintain the balance between heat loss and heat production in order to sustain body temperature within a normal range of 36.5°C - 37.5°C; this ability is limited in the new-born.

Hypothermia (cold stress, <36.5°C) and hyperthermia (>37.5°C) can have serious effects in neonates, increasing risk of morbidity and mortality, particularly in preterm infants. For example, hypothermia can lead to hypoglycaemia, respiratory distress, hypoxia, metabolic acidosis, necrotising enterocolitis and poor weight gain (McCall et al., 2018), increasing risk of unnecessary admissions of term infants to neonatal units (NHS Improvements, 2017).

The maintenance of the neural thermal environment is the ultimate aim of neonatal temperature control and management.

This document includes quick reference guides on the following:

- **Choice of Incubator/cot**
- **Delivery room algorithm**
- **The warm chain**
- **Hot cot flow chart**
- **Humidity flow chart and table**
- **Weaning to standard cot**

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QUICK REFERNCE GUIDES

Figure 1 Choice of Incubator/cot

Gestation/weight of infant	Type of Incubator/Cot according to gestation, weight and condition of infant					
<32 weeks gestation or <1500g	Double walled humidified incubator	Giraffe Omnibed (NDDH)				
32-36 weeks gestation or >1500g	Double walled humidified incubator	Giraffe Omnibed (NDDH)	Overhead heated incubator	Baby Therm	Hot Cot Consider normal cot >1600g	
>36 weeks gestation		Giraffe Omnibed (NDDH)	Overhead heated open incubator	Baby Therm	Hot Cot	Normal Cot

Figure 2 Delivery Room Algorithm

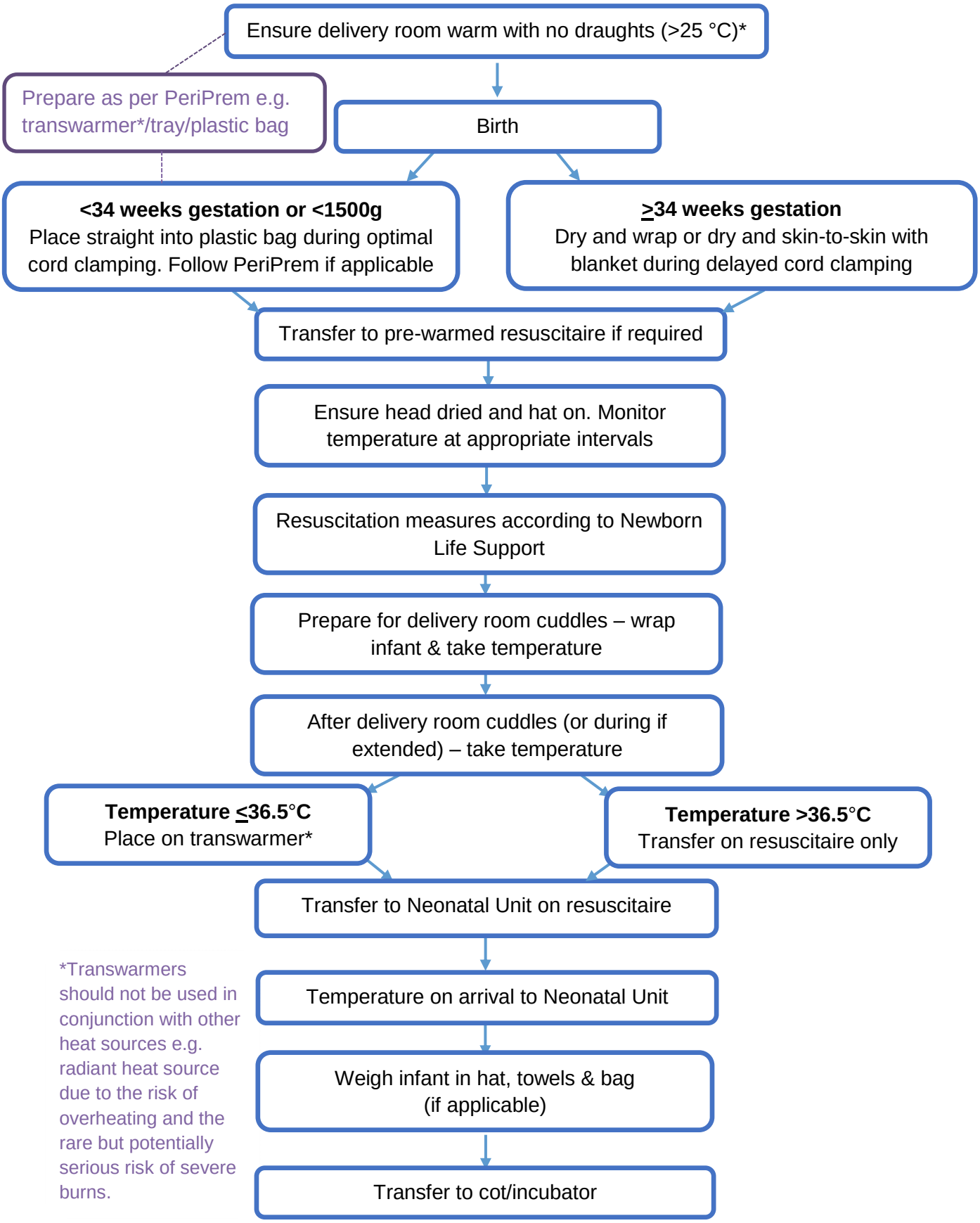


Figure 3 The Warm Chain

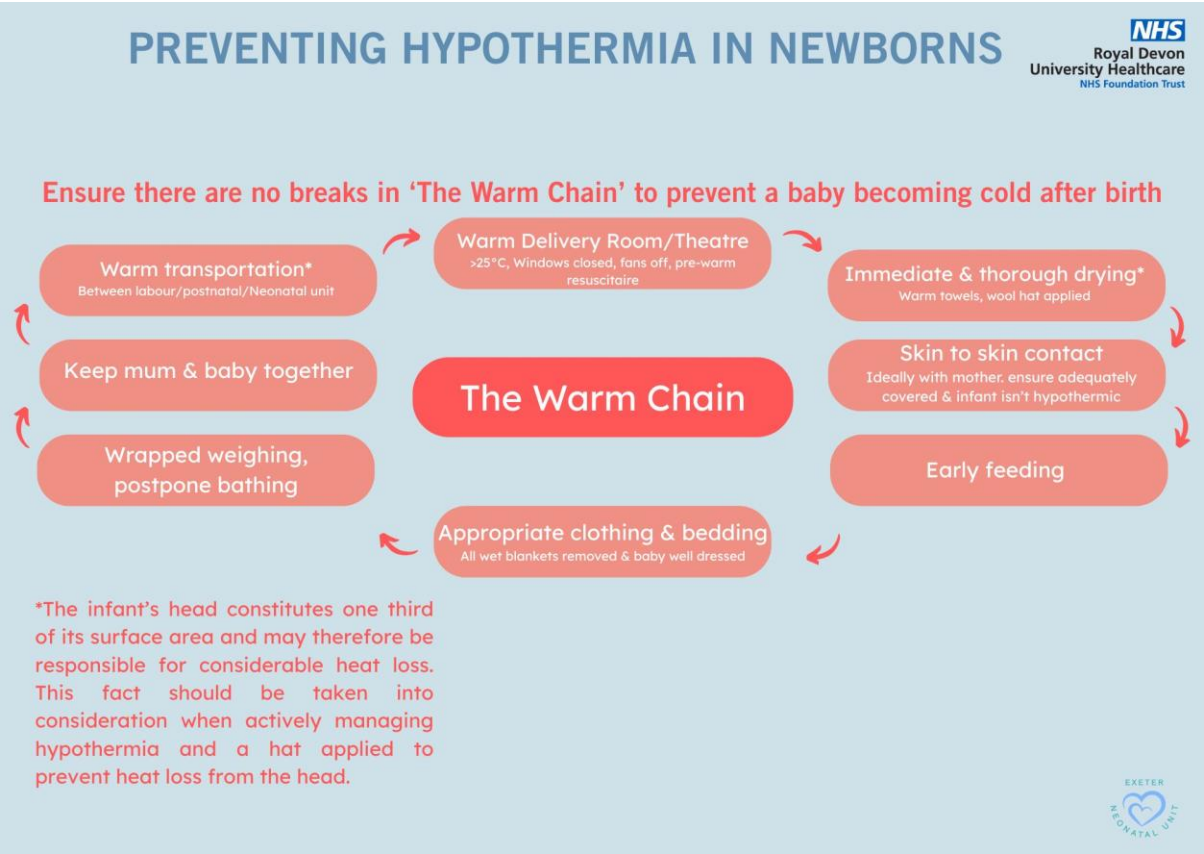


Figure 4 Hot Cot Flow Chart

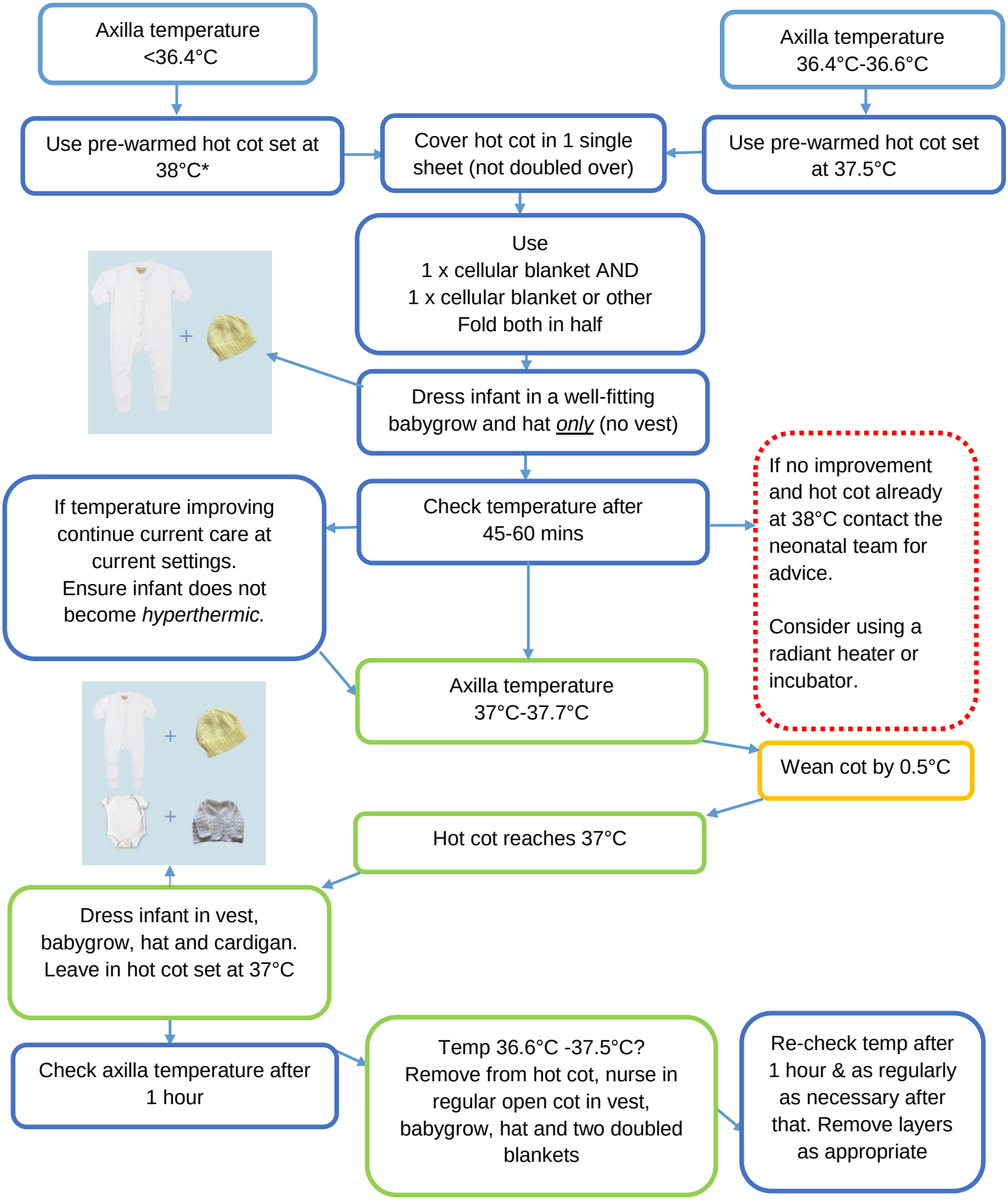


Figure 5 Humidity Flow Chart and Table

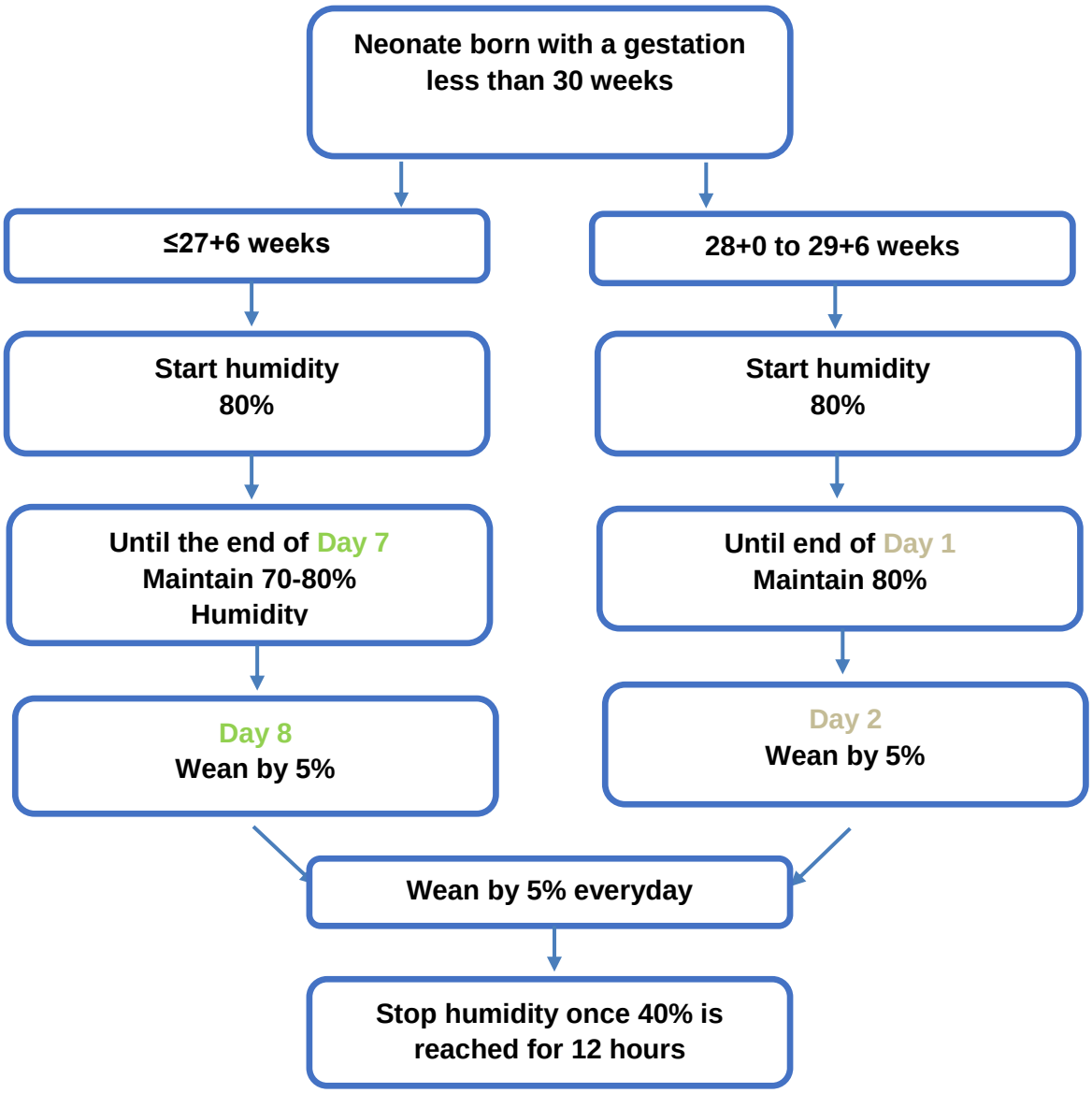


Figure 6. Weaning from an Incubator to Hot Cot or Standard Cot

Indicators for weaning are:

- Weight of >1600g

  (Weaning preterm infants from incubator to cot at a lower body weight of 1600 g has been found to accompany higher weight gain post-transfer compared with weaning at a higher body weight of 1800g (Razak, 2019))
- Medically stable condition with absence of further medical concern (i.e. no apnoeas or bradycardias requiring stimulation.
- Consecutive/consistent weight gain appropriate to age and gestation.
- Incubator temperature 28°C or less for 48 hours prior to transfer and baby maintaining a normal axilla temperature (36.5°C-37.5°C) with clothing/blanket/hat.

When ready for a cot

- Prior to transfer dress baby with well-fitting vest, sleepsuit, cardigan and hat. Preferably these have been warmed in the incubator before dressing the baby.
- When transferred into cot, cover with a cellular blanket plus 2 blankets doubled over.
- Once transferred axilla temp should be checked hourly for the first 4 hours.
- If within this time axilla temp is 36.2°C-36.5°C then consider transferring to a hot cot.
- If axilla temperature is less than 36.2°C then return to incubator.
- If after 4 hours axilla temperature is within normal range then recheck axilla temperature 4 hourly for the next 48 hours.
- Avoid bathing on day of transfer to the cot.

1. PURPOSE

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 The purpose of this document is to detail the process for the thermal care of neonates.

1.3 The policy applies to all staff responsible for the thermal care of neonates.

1.4 Implementation of this policy will ensure that:

- We create and maintain a neutral thermal environment for the baby
- We minimise the risks associated with hypo/ hyperthermia in the neonate

2. DEFINITIONS / ABBREVIATIONS

2.1 SCU - Special Care Unit at North Devon District Hospital

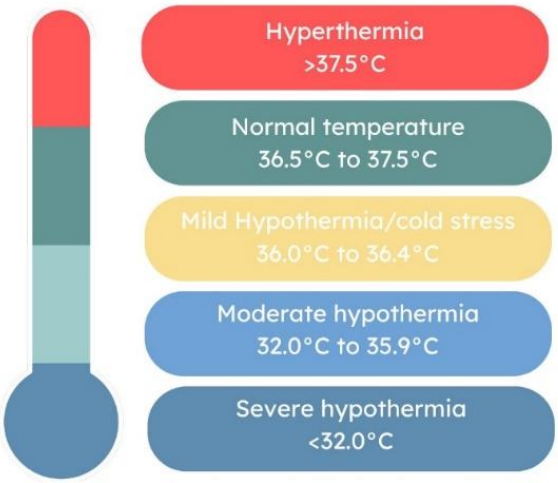
2.2 NNU – Neonatal Unit in Exeter (Eastern)

2.3 Neutral Thermal Environment – This is the temperature range where oxygen consumption and metabolic rate is at the minimum needed to maintain normal body temperature. It depends on birth weight, gestation and whether the baby is clothed. Each infant has his/her own individualized neutral thermal environment.

2.4 EH – Environmental Humidity – the concentration of water vapour in the air surrounding the neonate. Usually delivered inside an incubator.

2.5 TEWL – Trans-epidermal water loss – the evaporative loss of water from the immature epidermal layer of the premature neonate's skin.

2.6 Temperature ranges



Consider 36.8°C -37.2°C to be optimum

3. RESPONSIBILITIES

3.1 The Neonatal / Paediatric medical team are responsible for:

- Acting in accordance with this guideline
- Ensuring that risks associated with poor temperature regulation are minimised
- Keeping the Tertiary unit updated on condition of infant and delivering collaborative plan of care.

3.2 Midwifery and Neonatal nursing staff are responsible for:

- Keeping medical team aware of infant's condition
- Observing the baby for associated risks and taking appropriate action
- On-going monitoring and care of the baby

4. GENERAL PRINCIPLES OF THERMAL CARE OF THE NEONATE

Thermoregulation is the ability to maintain the balance between heat loss and heat production in order to sustain body temperature within a normal range of 36.5°C -37.5°C; this ability is limited in the new-born.

Hypothermia (cold stress, <36.5°C) and hyperthermia (>37.5°C) can have serious effects in neonates, increasing risk of morbidity and mortality, particularly in preterm infants. For example, hypothermia can lead to hypoglycaemia, respiratory distress, hypoxia, metabolic acidosis, necrotising enterocolitis and poor weight gain (McCall et al., 2018), increasing risk of unnecessary admissions of term infants to neonatal units (NHS Improvements, 2017).

The maintenance of the neural thermal environment is the ultimate aim of neonatal temperature control and management.

4.1 Pathophysiology

Premature and growth restricted infants are most at risk of hypothermia as they have less brown fat stores, poor vasomotor responses and less insulation. Their biochemical and hormonal response are immature and potentially inadequate. The main source of heat production in the neonate is non-shivering thermogenesis - the production of heat by metabolism.

Causes of hypothermia (cold stress):

- Prematurity
  - large surface area to body ratio
  - trans epidermal water loss (thin layer of skin)
  - less brown fat stores
  - poor vasomotor responses
  - less insulation.

- Environment

  Heat loss occurs when heat is transferred to the environment by:

  - *Evaporation* - when fluid evaporates from the skin and respiratory tract
  - *Conduction* - Loss of heat to cooler objects in direct contact with the skin
  - *Convection* - Heat loss to cooler surrounding air.
  - *Radiation* - Heat loss to cooler objects not in direct contact with the body.

4.2 Complications of hypothermia include:

- Increased glucose utilisation which may cause hypoglycaemia
- Increased oxygen consumption (hypoxia)
- Metabolic acidosis
- Pulmonary vasoconstriction
- Decreased surfactant production
- Disseminated intravascular coagulation
- Loss of weight or failure to gain weight

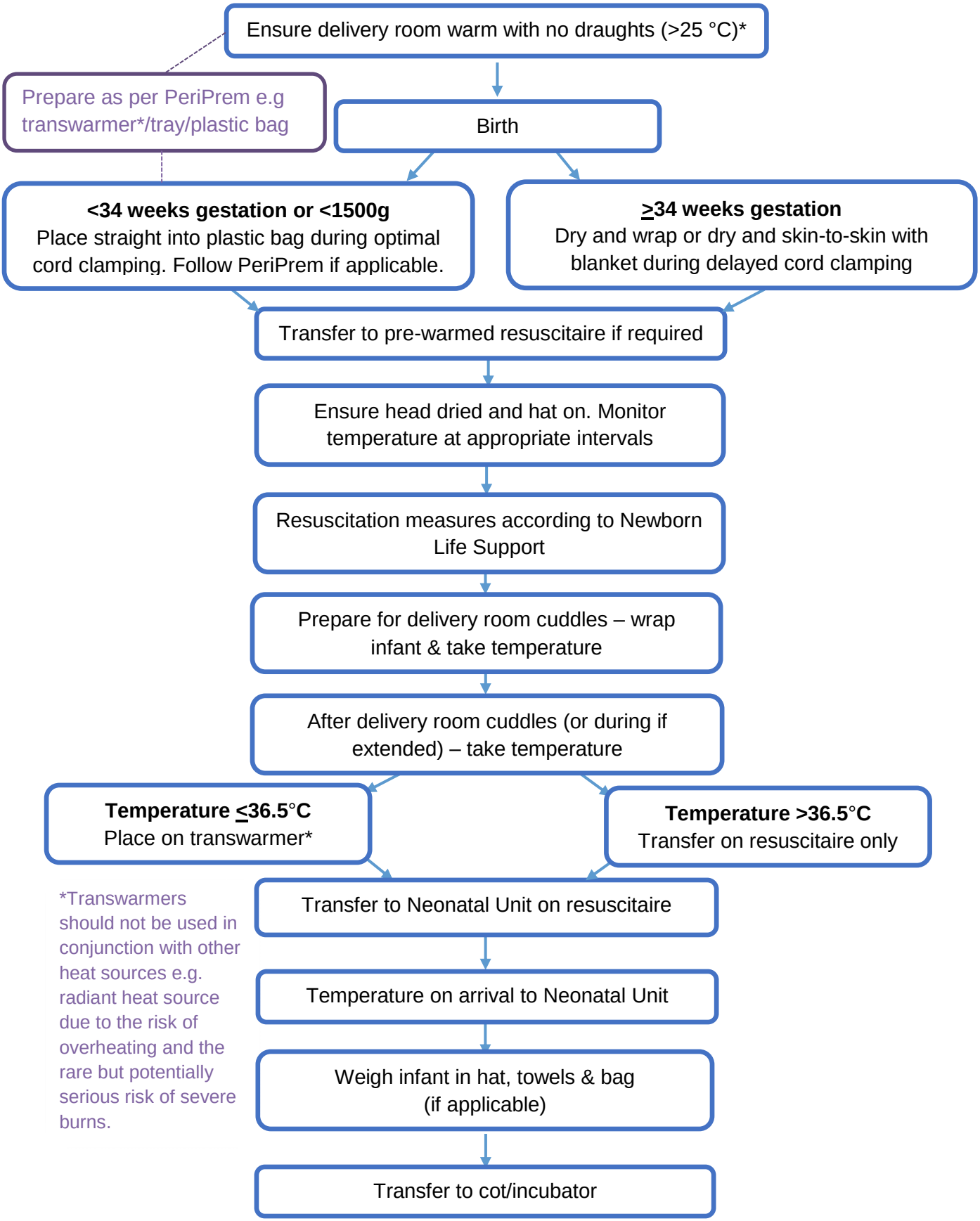
4.3 Signs of hypothermia include:

- Shallow breathing, apnoea
- Decreased activity and apparent lethargy
- Bradycardia or tachycardia
- Hypotonia with diminished reflexes
- Pale mottled skin – cool to touch, extremities
- Week suck, poor feeding
- Respiratory distress, tachypnoea
- Increased capillary refill time, i.e. > 2 seconds

5. METHODS OF MEASUREMENT OF TEMPERATURE IN NEONATES

- The axilla is the preferred site for measurement of temperature.
- The rectal temperature provides the core temperature and will only provide a late indication of cold stress and may cause trauma.
- Skin temperature may be continuously monitored by sensor or probe. The baby may lay on the probe if the mattress is not heated.
- If the infant is lying on a heated mattress then the probe should be well secured under the nappy (but not between the baby and the mattress) placed over the abdomen when supine or the back when prone avoiding bony prominences and excoriated areas.

6. THERMAL CARE IN THE DELIVERY ROOM



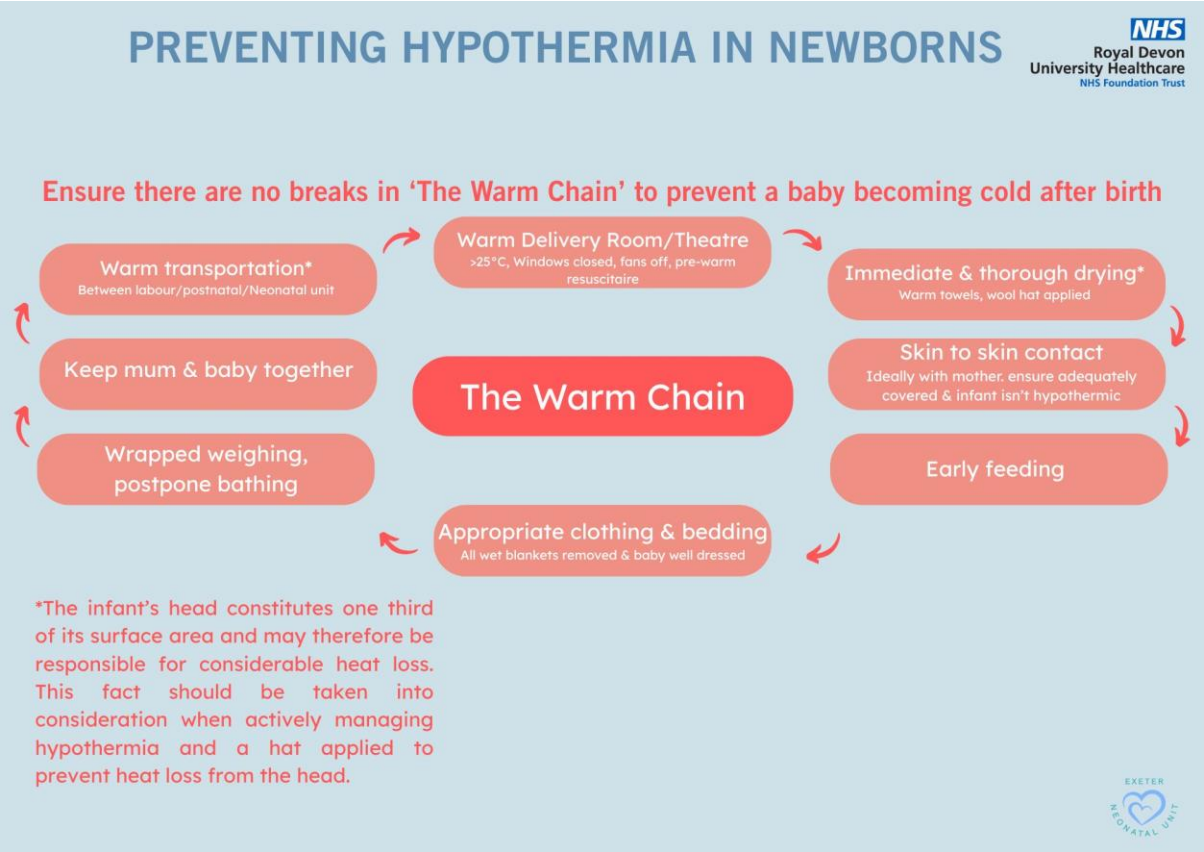
6.1 Monitoring the infant's temperature after birth

Within an hour of delivery record the infant's temperature. If normothermic recheck the infant's temperature with mother's observations and again at baby check.

If the infant is at risk of or showing signs that he may be hyper or hypothermic check this sooner and take appropriate action.

7. THE WARM CHAIN

These are a set of points derived by WHO to prevent hypothermia - most should be achievable. These should be considered for every baby and particularly those at risk.



8. SKIN TO SKIN

Prolonged skin to skin contact between mother and baby straight after delivery is associated with better thermoregulation of the newborn and therefore a decreased incidence of associated harmful side effects (NHS Improvement, 2017). Where medical attention is not necessary, skin to skin contact should be encouraged for at least 2 hours or until after the first feed to allow newborn thermoregulation (Unicef, 2017).

9. INCUBATORS

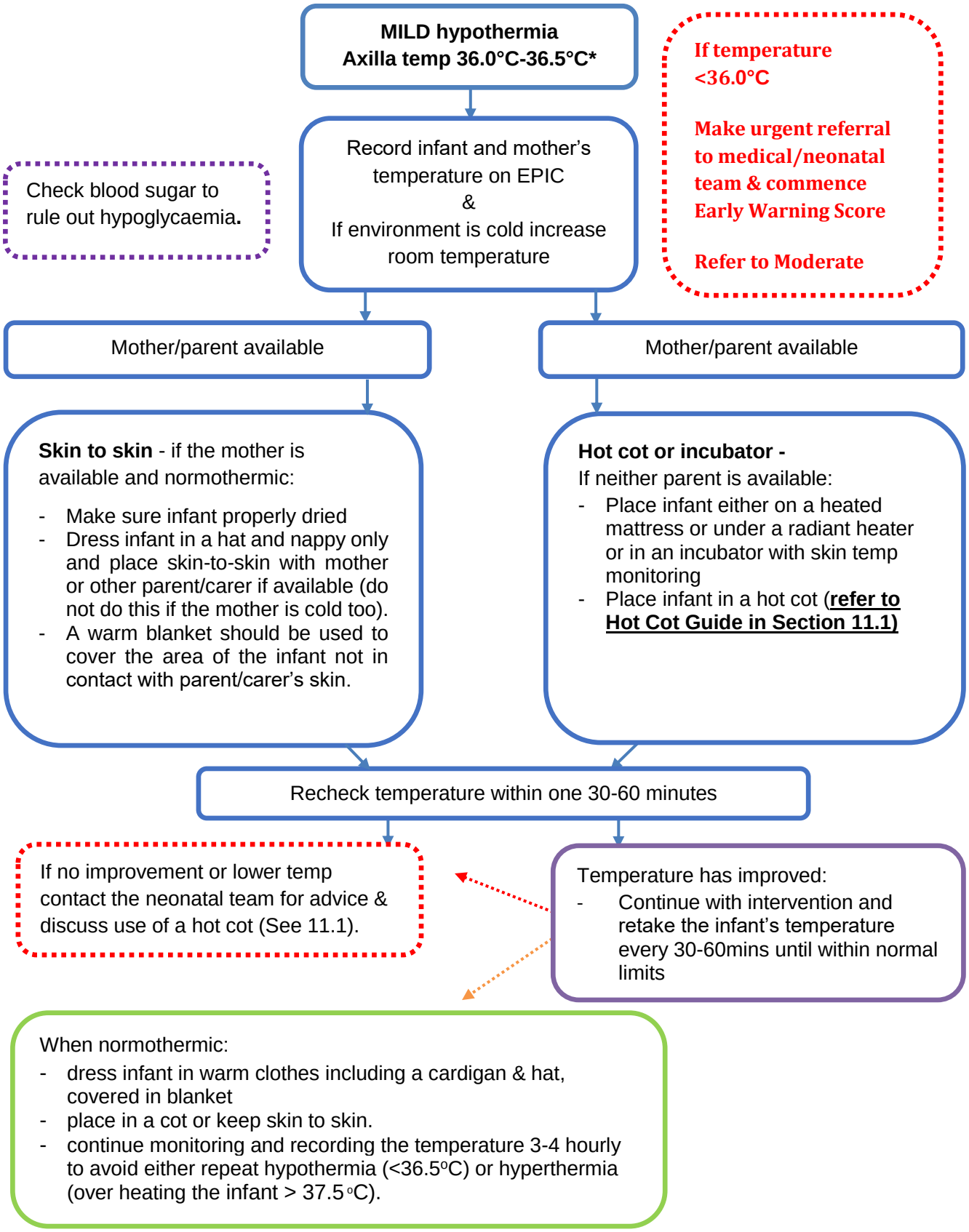
- Incubators are now specifically designed to minimise heat losses by radiation, convection, conduction and evaporation whilst allowing clear visibility and access to the neonate. Ambient temperature and humidity are easily controlled. Any neonate less than 1500g should be nursed within an enclosed incubator (Turnbull and Petty 2013)
- Before admission pre heat the incubator temperature according to the specific age and gestation and adjust incubator temperature according to individual response
- Alter set temperature according to the neonate's temperature and adjust by 0.5-1°C every 15-30 minutes, depending on the extent of temperature instability
- Do not leave a neonate inside a switched off incubator
- Do not expose an incubator to direct sunlight. Care and interventions, e.g. suction and nappy care, should be carried out via portholes, avoiding opening the side or roof completely
- The incubator should be changed every seven days and be recorded in the nursing documentation.

Suggested Incubator Temperatures for new admissions	
Weight (g)	**Incubator Temp. (degrees Celsius)**
500	37
1000	36
1500	35.5
2000	35
2500	34
3000	33

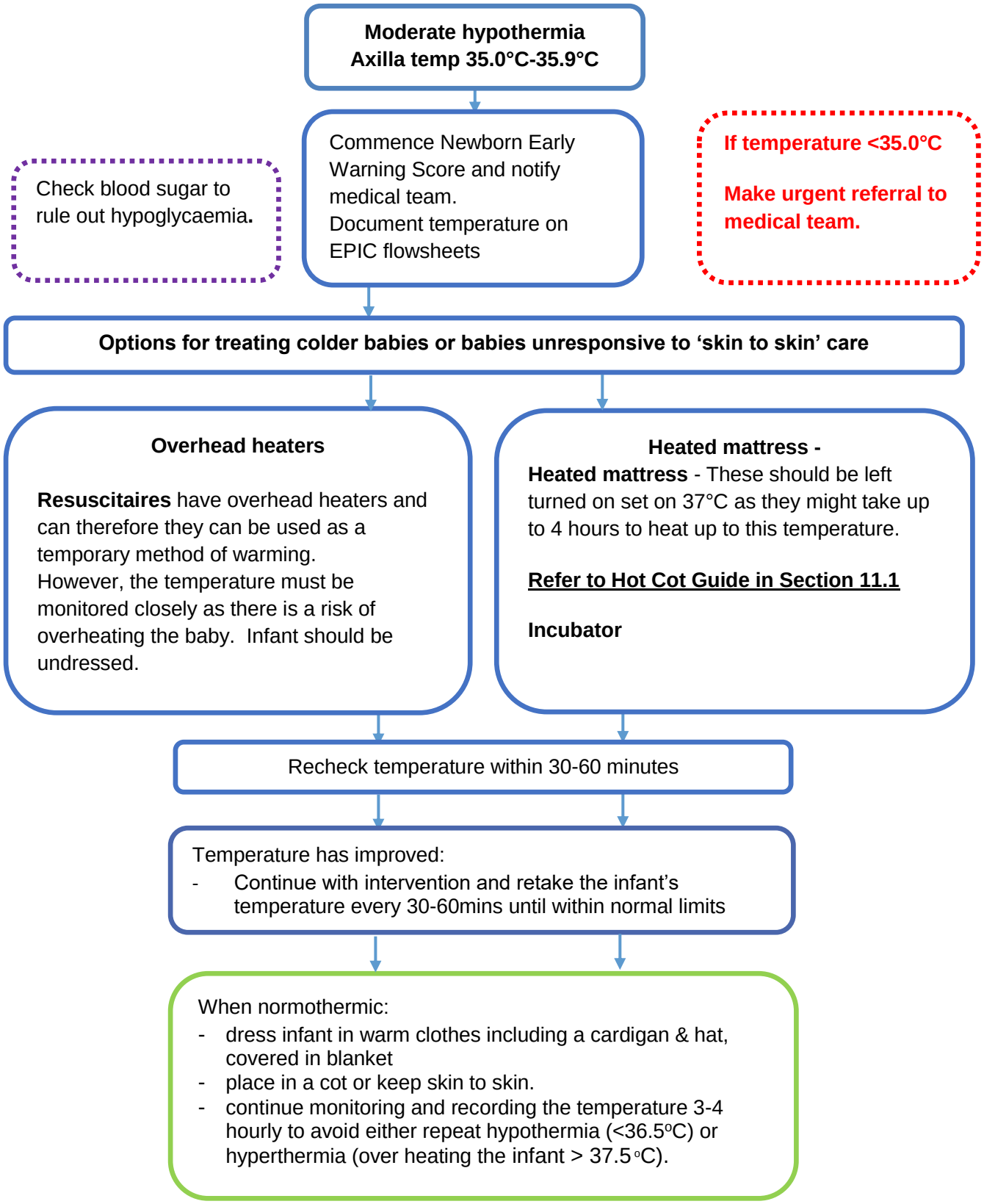
- Toys should not be placed in incubators, and only at the foot of the cot. Once the baby has been left to sleep toys should be removed from the sleeping environment.
- If toys are part of the baby's environment they should always be considered dirty and handled accordingly.
- Parents should be encouraged to take any toys home weekly and wash them in 60°c before returning them. If a cloth or 'snuggly' (that has been close to mum to absorb mother's smell) is left in the cot or incubator with baby this also should be washed regularly. This applies to all bedding etc. brought in by parents.

10. INTERVENTIONS FOR HYPOTHERMIA

10.1 Management of MILD hypothermia (Axilla temperature 36.0°C-36.5°C)



10.2 Moderate hypothermia (Axilla temperature 35.0°C to 35.9°C)



11. HOT COT

The hot cot is a simple device that is effective at preventing and treating hypothermia in the delivery suite or special/transitional care unit. It is as effective as an incubator for nursing the well preterm infant and enhances bonding with the parents as a physical wall is not interposed between them.

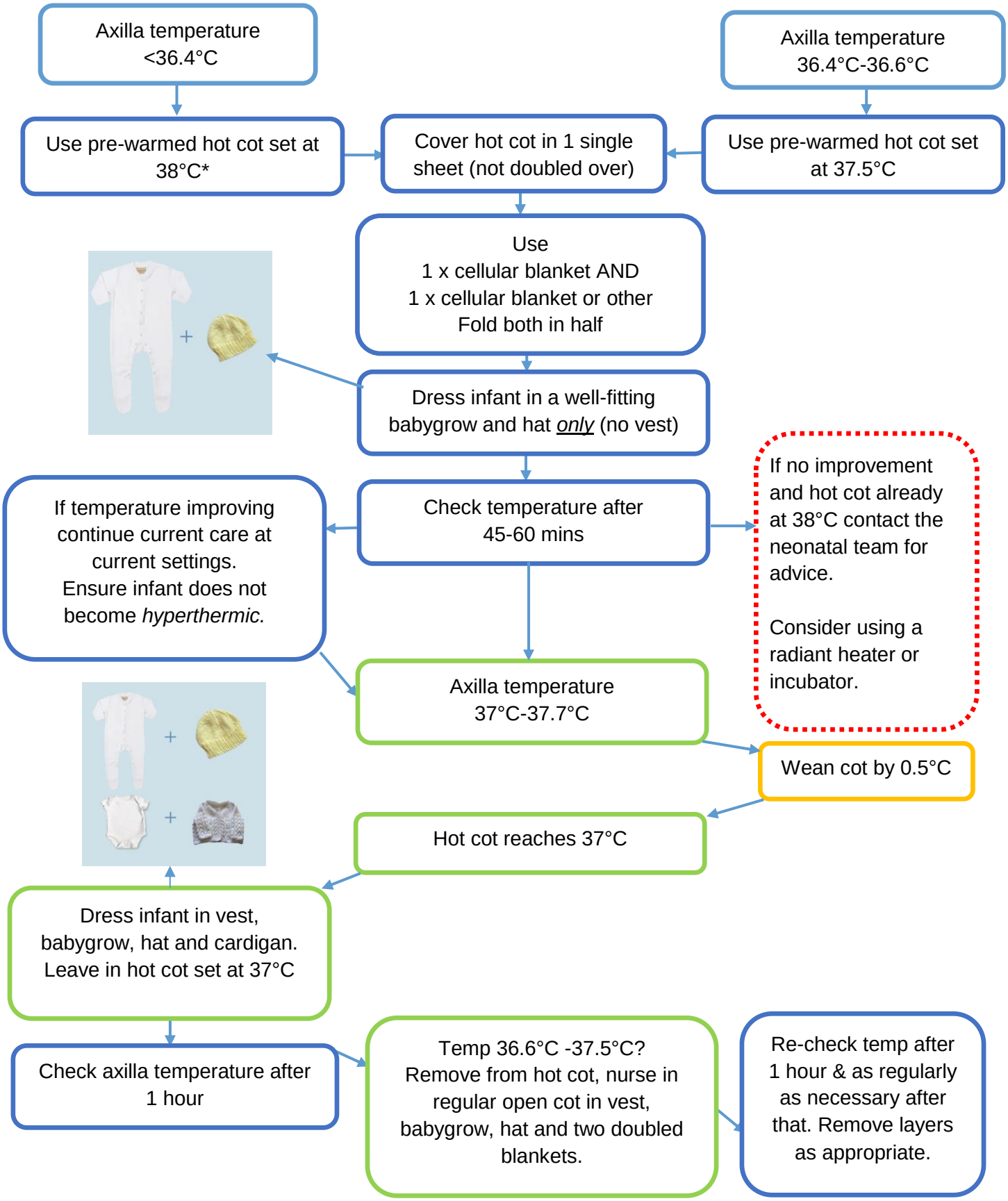
If used to warm near term infants, the hot cot should be removed once the temperature has stabilized as the infant may become too warm and start to sweat, inducing further hypothermia.

11.1 Criteria for transfer to a hot cot

Always inform the neonatal team if an infant is requiring a hot cot and they will assess the need for further review.

- ✓ Infant is born at less than 36 weeks and/or < 2kg
- ✓ Infant is receiving phototherapy and unable to maintain temperature within normal limits
- ✓ Babies who have had 2 consecutive low axilla temperatures of <36.6°C, despite intervention (skin to skin, extra clothing/hat/blanket, ensuring the room is not draughty)

11.2 Use of a hot cot



12. THERMAL CARE ON THE SCU/NNU & HUMIDITY

12.1 Classifications of core body temperature for newborns with actions in the Delivery Suite

TEMP	CLASSIFICATION	MANAGEMENT	ACTIONS
36.56-37.5^{0}C	Normal	Skin to skin, breast feed	Normal care*
36.0-36.5^{0}C	Mild Hypothermia or Cold Stress **Cause for concern**	Dress in thin layers, place in hot cot at 37^{0}C (cover not swaddle with warm blankets) until warm, monitor temperature	Consider informing Paediatrician and commencing NEWS observations
35 -35.9^{0}c	Moderate Hypothermia **(Danger - warm baby)**	As above	Inform Paediatrician commence NEWS obs. Complete Datix incident
32-34.9^{0}c	Moderate Hypothermia **(Danger - warm baby)**	Admission to SCU/NNU	Inform Paediatrician Complete Datix incident
<32^{0}c	Severe Hypothermia **(Outlook grave, skilled care urgently needed)**	Admission to SCU/NNU	Inform Paediatrician Complete Datix incident

(Adapted from WHO 1997 for management of newborns)

***Note - if baby's temperature is below 36.6°C on delivery suite or post natal ward, take and record mum's temperature and room temperature also.**

12.2 On Admission to SCU/NNU

- Ensure room infant is admitted to is 22°C-26°C (warmer according to prematurity of infant)
- Choose correct pre-warmed incubator/cot according to condition, gestation and weight of infant to best suit the baby to achieve and maintain a thermo-neutral environment (See quick reference guide).
- Always have warm hands before touching the baby.
- Ensure any surfaces the infant comes into contact with are pre warmed.
- Axilla temperature is checked and recorded (every 30 to 60 minutes) until the infant's temperature is within the accepted range (36.6°C to 37.3°C). The frequency

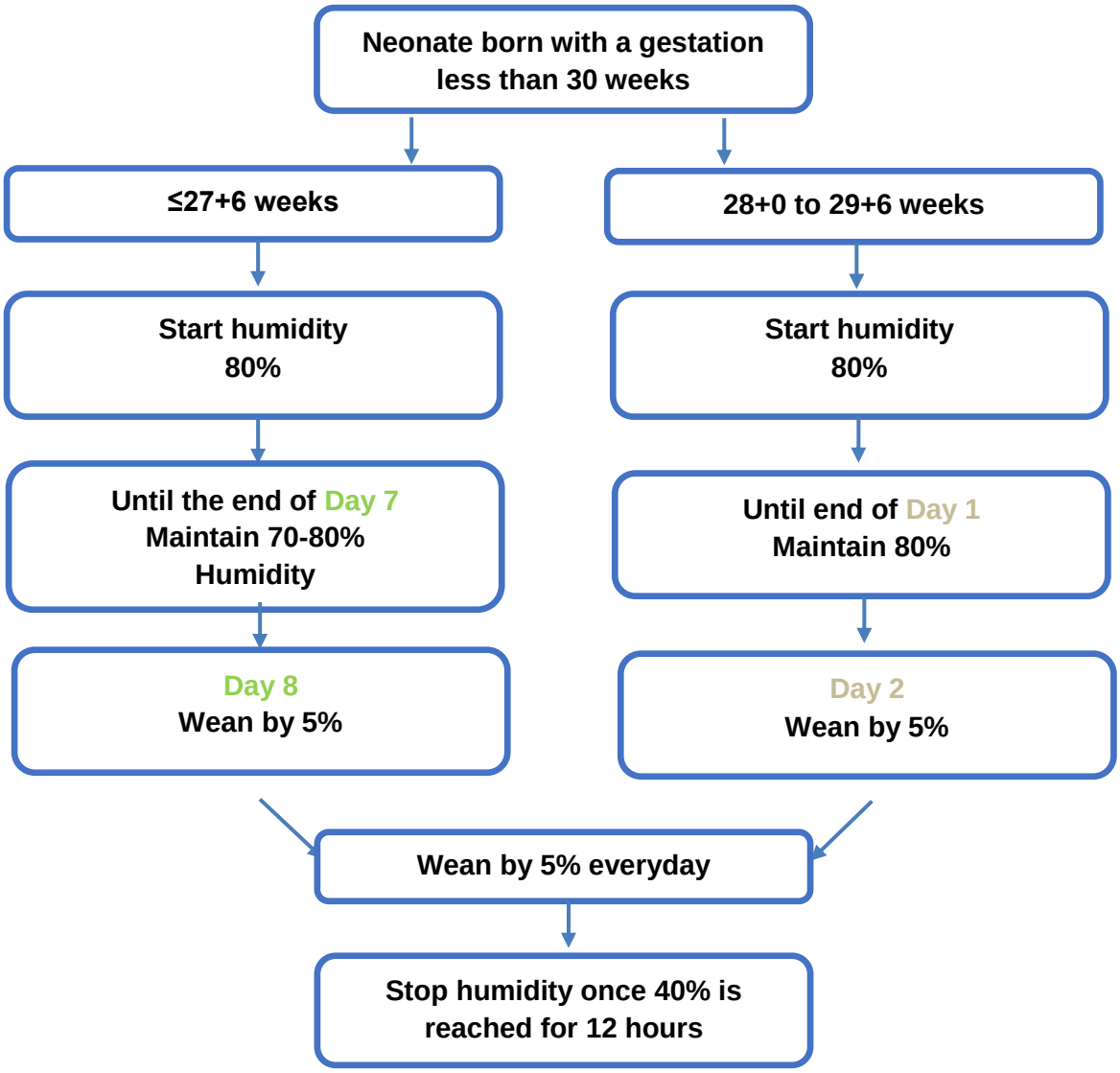
of measurement is then decreased according to the condition of the infant. (see Measurement of Temperature in Neonates above).

- If infant is nursed in an incubator attach a skin temperature probe (ensuring good contact) and record both skin and incubator temperature (including incubator set temp).
- Change environment according to temperature and condition of infant by adjusting humidity, temperature of incubator, overhead heater, mattress, hot cot or temperature of room to achieve a thermal neutral environment.
- Core and peripherals temperatures are monitored on the very sick intensive care neonate requiring retrieval or when an infant is passively cooled.
- If baby is hypo- or hyper- thermic, medical staff should be informed and axilla readings must be taken more frequently until the babies' temperature has normalized.
- If plastic bag is used remove only once procedures(e.g. placing a cannula) are finished and infant's temperature is within acceptable range.
- Observe minimal handling and keep incubator portholes closed.
- If a neonate undergoes any change of environment or increased exposure, e.g. procedures, phototherapy, new transfer to an incubator or cot, they will require checks for the first few hours until temperature is stable again.

12.3 Trans Epidermal Water Loss (TEWL) and incubator humidity

- The stratum corneum is a skin layer made up of corneocytes. This structure acts as a barrier which prevents water loss while also preventing entry of infectious organisms. The stratum corneum does not become mature until about 32 weeks gestational age.
- Prematurity of this barrier in neonates can lead to water loss, hypotension, hypothermia, electrolyte imbalance, infection, increased oxygen consumption, weight loss, and has been associated with significant morbidity.
- The earlier the infant is born, the higher the TEWL. Humidity provides a thermo-neutral environment which prevents thermal stress and can reduce the amount of heat that may be lost by evaporation.
- Adding heat and humidity to the environment of these infants can greatly reduce the amount of water loss, and assist with temperature regulation.
- After birth the skin of the preterm neonate is exposed to a gaseous environment and has been seen to mature rapidly, both in its epidermal structure and its barrier function. No matter their birth gestation, by 2-3 weeks of age the neonate's skin functions like that of a term baby.
- Nursing neonates in humidity has been shown to slow the natural process of epidermal barrier formation. Therefore, optimal incubator humidity levels and duration are key. Infants of different gestations have different requirements - all neonates will require a gradual reduction in incubator humidity as TEWL levels reduce and a transition is made from the artificially humidified environment to 'normal' room humidity.

12.4 Humidity guidance



12.4.1 Equipment

- Incubator – Drager Babyleo TN500 or equivalent
- Humidity line – Drager 'Transition set'
- Bag of sterile water for irrigation

12.4.2 Setting up humidity

- Run the sterile water through the humidity line and attach to the incubator (Sterile water is used, not tap water, to prevent colonisation by bacteria).
- Set incubator temperature to 35˚C, and humidity to 80%
- The neonate will have been delivered into a polyethylene bag (as per 'Thermoregulation Algorithm for delivery') - this should remain in place until the set percentage of humidity is reached within the incubator
- Regularly check and monitor the neonate's temperature throughout the admission process when humidity is commenced.

27+6 weeks and below		28+0 to 29+6 weeks gestation	
Start 80% humidity		Start 80% humidity	
Day 0-7	Maintain 70-80% humidity	Day 0-1	Maintain 80% humidity
Day 8	Wean to 65%	Day 2	75%
Day 9	60%	Day 3	70%
Day 10	55%	Day 4	65%
Day 11	50%	Day 5	60%
Day 12	45%	Day 6	55%
Day 13	40	Day 7	50%
Stop once 40% for at least 12 hours		Day 8	45%
		Day 9	40%
		Stop once 40% for at least 12 hours	

12.4.3 Nursing care

- Aim to maintain the neonate's temperature 36.8˚C - 37.2˚C.
- The neonate should have their skin temperature monitored continuously with a temperature probe.
- Change the probe's position with cares.
- Check the axilla temperature with cares to ensure the probe's accuracy.
- Urine output should be measured and recorded with cares. There is very minimal absorption of humidity into the nappy, so the values should be accurate.
- Try to keep incubator doors shut/sides up as much as possible to help retain humidity.
- To ensure effective humidity the neonate should be nursed with the smallest appropriately sized nappy, and no clothing. Bedding can be used to nest and position, but blankets should not be placed over the neonate.
- Use a thick incubator cover to insulate the incubator and reduce condensation.

- It is recommended that soft toys are removed from incubators, this is because bacteria on soft toys may proliferate in the extra warm and humid environment. Snugglies with parental scent may be used instead.
- The humid environment increases the risk of skin infections. The neonate's skin should be observed regularly for signs of infection.
- Phototherapy can be used for neonates receiving humidity, however carers should be aware of temperature fluctuations caused by the heat from the phototherapy.
- Skin to skin is still recommended for neonates being nursed in humidity.
- Care for every baby should be individualised - if carers feel that a neonate should not follow the humidification protocol outlined this decision should always be made in discussion with the medical/nursing team.
- **Hourly**: document temperature probe reading, humidity setting and incubator temperature.
- **With cares**: document condition of skin, axilla temperature, fluid balance.

12.4.4 Family-Centered Care

- Explain to parents the purpose of humidity, and keep them updated about progress and weaning.
- Inform parents why toys, clothing, and blankets should not be used alongside humidity.
- Encourage holding - evidence indicates that preterm infants were able to maintain stable temperatures while outside of the humidified incubator during skin to skin.

12.4.5 Cleaning

- During use, if the incubator needs to be cleaned on the inside it should ideally be changed. If this is not possible, then plain water only can be used to clean the internal surfaces.
- All incubators should be changed after 7 days. The Drager Giraffe should be changed every 14 days.
- Once humidification has been stopped change the incubator to prevent microbial build up.
- Before cleaning put a 'Humidity used' sign on the incubator so NNU Assistants know whether humidification has been used.
- Due to becoming easily contaminated with bacteria toys should not be placed in incubators.

13. SUSPECTED HYPOXIC ISCHAEMIC ENCEPHALOPATHY (HIE)

Hypothermia is currently being used as a method of cerebral protection in the infant that has been compromised by a hypoxic insult. Term infants with suspected Hypoxic Ischaemic Encephalopathy (HIE) may be considered for treatment with cooling. Please refer to the South West Neonatal Network Therapeutic Hypothermia Care Pathway Guideline.

14. RE-WARMING A COLD INFANT ON SCU/NNU

- Determine the cause of low temperature and treat appropriately. Hypothermia may be due to environmental conditions or an underlying disease process.
- If Infant is nursed in an incubator:
  - Maintain the ambient temperature 1 to 1.5°C higher that the infant's temperature
  - Check humidity is correct for gestation
  - Temperature should be checked every 30 minutes to assess the effectiveness on the intervention, use a skin probe to monitor continuous skin temperature.
  - Increase air temperature hourly by 1°C until the infant's temperature is stable in the normal range. *(Rapid re-warming can provoke apnoeic episode and hypotension or even seizures)*
  - Remove plastic wraps, caps, etc. whilst re-warming as these can trap cold air.
  - Reduce the temperature slowly as the baby warms up
  - Continuously monitor oxygen saturations and administer oxygen as indicated.
  - Check blood sugars regularly to monitor any hypoglycaemia.
- If infant is nursed in a cot consider transferring to:
  - a humidified incubator (see above)
  - an overhead heated open incubator controlled by skin temperature sensor
  - babytherm/resuscitaire
  - a hot cot/warm mattress (See section 11)

15. WEANING FROM AN INCUBATOR TO HOT COT OR STANDARD COT

Indicators for weaning are:

- Weight of >1600g
  (Weaning preterm infants from incubator to cot at a lower body weight of 1600 g has been found to accompany higher weight gain post-transfer compared with weaning at a higher body weight of 1800g (Razak, 2019))
- Medically stable condition with absence of further medical concern (i.e. no apnoeas or bradycardias requiring stimulation.
- Consecutive/consistent weight gain appropriate to age and gestation.
- Incubator temperature 28°C or less for 48 hours prior to transfer and baby maintaining a normal axilla temperature (36.5°C-37.5°C) with clothing/blanket/hat.

When ready for a cot

- Prior to transfer dress baby with well-fitting vest, sleepsuit, cardigan and hat. Preferably these have been warmed in the incubator before dressing the baby.
- When transferred into cot, cover with a cellular blanket plus 2 blankets doubled over.
- Once transferred axilla temp should be checked hourly for the first 4 hours.
- If within this time axilla temp is 36.2°C-36.5°C then consider transferring to a hot cot.
- If axilla temperature is less than 36.2°C then return to incubator.
- If after 4 hours axilla temperature is within normal range then recheck axilla temperature 4 hourly for the next 48 hours.
- Avoid bathing on day of transfer to the cot.

16. THERMAL CARE FOR BABIES RECEIVING PHOTOTHERAPY

Phototherapy can be delivered by several different methods and it is necessary to check the baby's temperatures at regular intervals.

Axilla temperature should be checked and recorded 30 minutes after phototherapy has commenced and then hourly for the next 4 hours.

If temperature has been stable axilla temperature should then be checked and recorded every 4 hours until phototherapy stops.

When phototherapy has been stopped axilla temperature should be checked after 30 minutes. If temperature within normal limits check again after another hour and if remains within normal limits baby can return to normal care.

17. HYPERTHERMIA

Hyperthermia rarely occurs in a neonate by means of anything other than an external source. In an actual febrile state, the rectal temperature should be higher than a distal temperature. The reverse is true with overheating.

17.1 Risk factors for hyperthermia include:

- Over enthusiastic re-warming
- Warming and phototherapy lamps
- Sepsis in the older infant
- Neonatal abstinence syndrome
- Prostaglandin therapy
- Brain damage to areas of thermoregulatory control
- Thyrotoxicosis
- Maternal pyrexia

17.2 Signs and symptoms of hyperthermia

- Tachypnoea and apnoea
- Tachycardia
- Increased oxygen requirements
- Metabolic acidosis
- Hypotension
- Lethargy, hypertonia & extended posture
- Irritability
- Flushing

17.3 Management of Hyperthermia

- Document full set of NEWS/ Special Care observations
- Consider any environmental factors e.g. radiant heat source, too many blankets
- Remove excess layers and clothing
- Check environmental temperature and reduce by 0.5°C at 30-60 minute intervals where possible
- If under a Babytherm or resuscitaire, turn radiant heater off and choose the extended lower range (<35°C).Turn the temperature down by 0.5° at 30-60 minute intervals (Parsons, 2017)
- Discuss with medical team if temperature over 37.3°c and rising and/or does not reduce after adjusting environment
- If the cause is not thought to be environmental, consider infection/ sepsis (please see Sepsis Management Guidelines (early and late onset) for Neonates).

18. MONITORING COMPLIANCE WITH AND THE EFFECTIVENESS OF THE GUIDELINE

Key performance indicators comprise:

- Nice Neonatal Quality Standards
- NHS Toolkit for High Quality Neonatal Services
- National Neonatal Audit Programme
- NHS Standard Contract for Neonatal Critical Care as their Key Performance indicators on which to base care.
- ATAIN – NHS Improvement - Reducing harm leading to avoidable admission of full-term babies into neonatal units

19. PROCESS FOR IMPLEMENTATION AND MONITORING COMPLIANCE AND EFFECTIVENESS

- Staff are informed of revised documentation. There is an expectation that staff are responsible to keep updated on any improvements to practice and deliver care accordingly. Data is collected by use of Badger data base and can be used to generate output for clinical and operational benchmarking.
- Data is used for the following purposes, BAPM neonatal dataset (2012), Neonatal Critical Care Minimum Data Set, National Neonatal Dashboard, National Neonatal Audit Programme, Mothers and Babies Reducing Risk through Audits and Confidential Enquiries (MBBRACE) Dataset and South West Neonatal Network Dashboard.
- Admission temperature data is collected and monitored by the South West Neonatal Network dashboard and by the Neonatal Critical Care dashboard.
- Non-adherence to the guideline and moderate to severe hypothermia is reported by use of the Datix system. Incidents are monitored and reviewed by the maternity and neonatal governance teams and action plans made if required.
- Further discussion and reviews occur at Directorate meetings, Maternity/Neonatal/Paediatric Governance meetings and unit meetings. Learning and action plans are cascaded at these meetings and improvements implemented. Key findings and learning points will be disseminated to relevant staff

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